

1. DETERMINATION OF RADIUS OF CURVATURE OF PLANO CONVEX LENS- NEWTON'S RINGS

Aim: To determine the radius of curvature of a plano-convex lens using a monochromatic source of light of known wavelength.

Apparatus:

Sodium vapour lamp, plano-convex lens of large focal length, a plane glass plate, reflecting plane glass plate, a dark paper, travelling microscope and retard stand.

Formula:

The Radius of curvature 'R' for a given plano convex lens is determined by using

$$R = \frac{d_n^2 - d_m^2}{4\lambda(n - m)} \text{ cm} \quad \text{----- (1)}$$

where d_n = diameter of n^{th} ring.

d_m = diameter of m^{th} ring.

λ = Wave length of incident monochromatic light.

Description:

A black paper is placed on the base of the travelling microscope over which a thin glass plate 'P' is placed. Upon this a plano convex lens of radius of curvature to be determined is placed. A parallel beam of monochromatic light coming from sodium vapour lamp is made to incident at an angle of 45° as shown in fig. As a result of interference of glass plate a concentric ring of alternate dark and bright fringes are seen through microscope which are called as Newton rings. The eyepiece of microscope is adjusted so that rings are viewed uniformly and the glass plate is also adjusted slightly.

$$\text{Least Count} = \frac{\text{one Main scale Division}}{\text{Total number of vernier scale Divisions (N)}}$$

$$1. \text{ M.S. D} = 0.05\text{cm.}$$

$$N = 50$$

$$L.C = \frac{1\text{MSD}}{N} = \frac{0.05\text{cm}}{50} = 0.001\text{cm}$$

Procedure:

The least count of travelling microscope is determined. The point of intersection of the iron wire in microscope is brought to the ring system. The microscope is moved to one side counting the number of rings. After passing beyond 25^{th} ring the direction of microscope is reversed and by coinciding the cross wire with 20^{th} , 15^{th} , 5^{nd} dark rings are noted. The microscope is moved in the same direction and readings of the corresponding 5^{nd} , 10^{th} ,, 25^{th} dark ring on the right side of dark region are noted and observations are tabulated. The wavelength of monochromatic light of sodium vapour lamp is taken from standard tables.

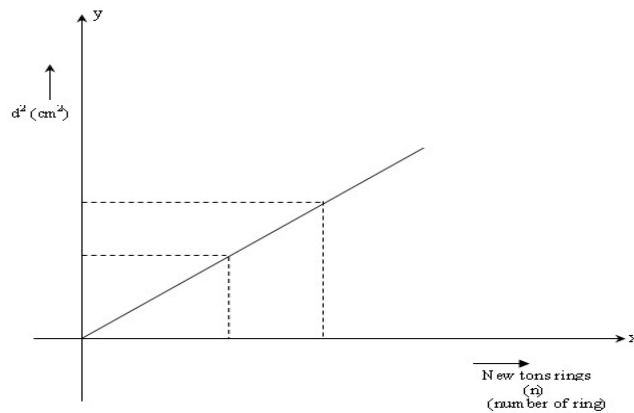
Graph:

A graph is drawn with no. of rings as abscissa (x – axis) and the square of diameter of the ring as ordinate (y-axis). The graph will be a straight line. From the graph the values of d_1^2 & d_2^2 corresponding to the two number of rings n_1 & n_2 are noted.

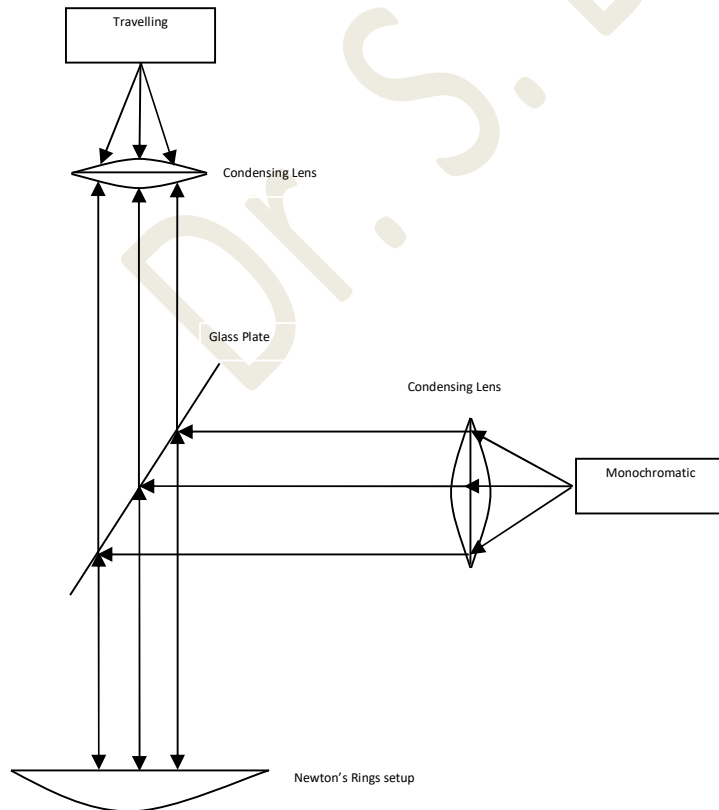
Using these values on equation (1) the wavelength of the source ' λ ' is calculated.

Then the wavelength of the light radiation is

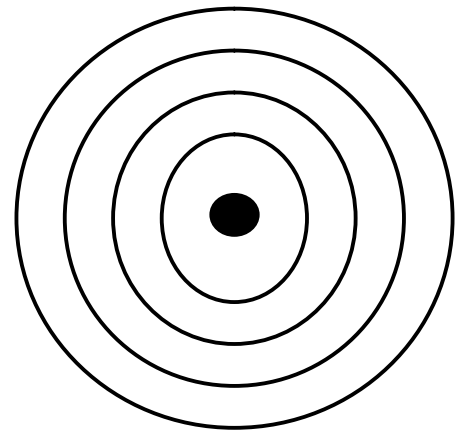
$$\lambda = \frac{d_n^2 - d_m^2}{4R(n - m)} \text{ cm}$$



Diagrammatic representation:



Interference pattern:



Tabular form:

S.No	No. of rings (n)	Microscope Readings								Diameter d=x~y cm	d^2 (cm) ²	$d_n^2 - d_m^2$ (cm) ²
		Left				Right						
		M.S.R (a) (cm)	V.C	V.C X L.C (b) (cm)	Total (a+b) x (cm)	M.S.R (a) (cm)	V.C	V.C X L.C (b) (cm)	Total (a+b) y (cm)			
1	5											
2	10											
3	15											
4	20											
5	25											

Calculation:

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Precautions:

- Glass plate and lens should be cleaned thoroughly.
- Lens used should be of large radius of curvature
- The microscope eye lens and reflecting glass plate are adjusted properly
- The rings formed are not to be disturbed till the experiment is completed.

Result:

1. Radius of curvature of given plano convex lens from table, $R =$ _____ cm.
2. Radius of curvature of given plano convex lens from graph, $R =$ _____ cm.